

Project Submission — PawPrep: Controlled Veterinary Illustrations

Course project | Controlled Image Generation with Stable Diffusion

GitHub: <https://github.com/reddy-nithin/stable-diffusion>

Submission tag: `v1.0-submission`

Slide Deck (10 Slides)

[Open presentation](#)

Slide	Rubric item covered
1	Scenario description
2	Dataset
3	Methodology (2×2 ablation design)
4	Pipeline (Stable Diffusion + ControlNet)
5	Prompt design
6	Control strategy
7	Tools & technologies
8	Results + Evaluation (real metric values)
9	Demo video link + GitHub link
10	Findings, limitations, AI tools disclosure

Demo Video (90 seconds)

[Watch demo](#)

Covers:

- System overview — PawPrep UI walkthrough
 - Input → prompt → output: Shiba Inu + grooming + ControlNet ON
 - Comparison: same breed/seed with ControlNet OFF vs ON
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GitHub Repository

<https://github.com/reddy-nithin/stable-diffusion>

Requirement	Location
Code	src/ — pipelines, prompts, evaluation, Gradio app
README	README.md — quickstart, pipeline diagram, sample grid
Dataset description	README.md → Dataset section; configs/taxonomy.yaml
Prompt strategy	docs/PROMPT_STRATEGY.md
Sample outputs	samples/ — 8 curated PNGs (7× cell D + 1× cell A for comparison)
Tools & libraries	README.md → Evaluation table; requirements.txt

Working Pipeline

Scenario

Veterinary clinics and pet-care educators need visual aids for common care situations (cone collar, bandaged paw, grooming, dental check, etc.) without expensive photography. PawPrep takes structured inputs — animal_type, breed, condition, environment — and produces consistent, breed-accurate educational illustrations.

Dataset

Oxford-IIIT Pet Dataset — 37 breeds (~7,400 images) with trimap segmentation masks. Loaded via torchvision.datasets.OxfordIIITPet. Trimaps converted to ADE20K seg maps for ControlNet conditioning.

Pipeline

```
Oxford trimap → seg map / canny edges ■■■
■■■ SD 1.5 + ControlNet → image + metadata.json
taxonomy.yaml + prompts.yaml ■■■■■■■■■■■■■■
```

- Ablation cells:
- **A** — naive prompt, no ControlNet (baseline)
 - **B** — naive prompt + ControlNet
 - **C** — structured prompt, no ControlNet
 - **D** — structured prompt + ControlNet

Key files: src/pipelines/baseline.py, src/pipelines/controlnet.py, src/generation/runner.py

Controlled Generation

Control mechanism: ControlNet-seg (lillyasviel/control_v11p_sd15_seg)

Trimap foreground → ADE20K animal class color (4, 200, 4) → ControlNet conditioning signal.

Conditioning strength 0.55, guidance end 0.6 — locks pose in early diffusion steps, releases in later steps so prompt drives breed details.

Canny fallback (l1lyasviel/control_v11p_sd15_canny) available when seg-map signal is weak.

Full write-up: docs/PROMPT_STRATEGY.md → Section 4 (Ethics Guard) and src/pipelines/controlnet.py

Evaluation and Analysis

Metrics

Metric	Model	Measures
CLIPScore	openai/clip-vit-base-patch32	Prompt-image alignment
Breed Identity	facebookresearch/dinov2 (DINOv2 ViT-B/14)	Visual breed fidelity vs Oxford reference
Condition Consistency	DINOv2 cosine across seeds	Semantic stability
LPIPS Diversity	torchmetrics	Output variety across seeds

Evaluation code: src/evaluation/ | Full run: notebooks/04_evaluation.ipynb

Results

Cell	Prompt	ControlNet	CLIPScore ↑	Breed Identity ↑	Consistency ↑	LPIPS Diversity ↑
A	Naive	Off	0.2960 ± 0.0285	0.5044	0.5926	0.6676
B	Naive	On	0.2966 ± 0.0202	0.6023	0.7419	0.5294
C	Structured	Off	0.3016 ± 0.0283	0.3080	0.5513	0.6864
D	Structured	On	0.3069 ± 0.0220	0.4375	0.6847	0.5444

Findings and Insights

Prompt effectiveness (C vs A)

Structured prompts improve CLIPScore (+0.0056, +1.9%) — detailed slot-filled prompts with style anchor and condition clauses produce better text-image alignment.

However, structured prompting *alone* reduces breed identity (0.3080 vs 0.5044) — detailed style/condition clauses dilute the breed signal without shape conditioning to anchor it.

Control vs diversity trade-off (B vs A, D vs C)

ControlNet dramatically improves breed identity (B: +0.0979 over A) and condition consistency (B: +0.1493 over A). Trade-off: diversity drops (LPIPS 0.67 → 0.53), as shape conditioning anchors the pose and reduces seed variance. All cells remain above LPIPS 0.52 — no mode collapse observed.

Diffusion limitations

- SD 1.5 fur texture quality below SDXL — over-smooth coat rendering across all cells
- Single-class seg maps (foreground only) weaker than full-scene semantic conditioning
- Rare conditions (dental check, vaccination) underrepresented in SD 1.5 training data — condition detail sometimes absent from output

Success and failure cases

Success: samples/07_shiba_grooming_D.png — correct orange/white Shiba coat, groomer's hand clearly visible.

Failure comparison: samples/02_samoyed_cone_clinic_A.png (Cell A) vs samples/01_samoyed_cone_clinic_D.png (Cell D) — same seed, A shows no cone and loose anatomy.

Full failure audit: docs/ETHICS.md → Failure Case Audit section

AI Tools Disclosure

Tool	Use
Claude (Anthropic)	Code generation, prompt strategy design, documentation drafting
Stable Diffusion v1.5	Image generation backbone — CreativeML Open RAIL-M license
ControlNet	Shape conditioning — Apache 2.0 license
Oxford-IIIT Pet Dataset	Training reference and ControlNet conditioning — CC BY 4.0
DINOv2	Breed identity evaluation — Apache 2.0
CLIP	Prompt-image alignment evaluation — MIT

All generated code was reviewed and tested by the author before use.
No model outputs were used as ground truth — all evaluation metrics are algorithmic.
Full statement: docs/ETHICS.md

Quick Reference

Deliverable	Link / Location
Slide deck	Google Slides
Demo video	Google Drive
GitHub repo	reddy-nithin/stable-diffusion
Sample outputs	<code>samples/</code> in repo
Evaluation notebook	<code>notebooks/04_evaluation.ipynb</code>
Prompt strategy	<code>docs/PROMPT_STRATEGY.md</code>
Ethics statement	<code>docs/ETHICS.md</code>
Colab quickstart	<code>README.md</code> → Quickstart section